



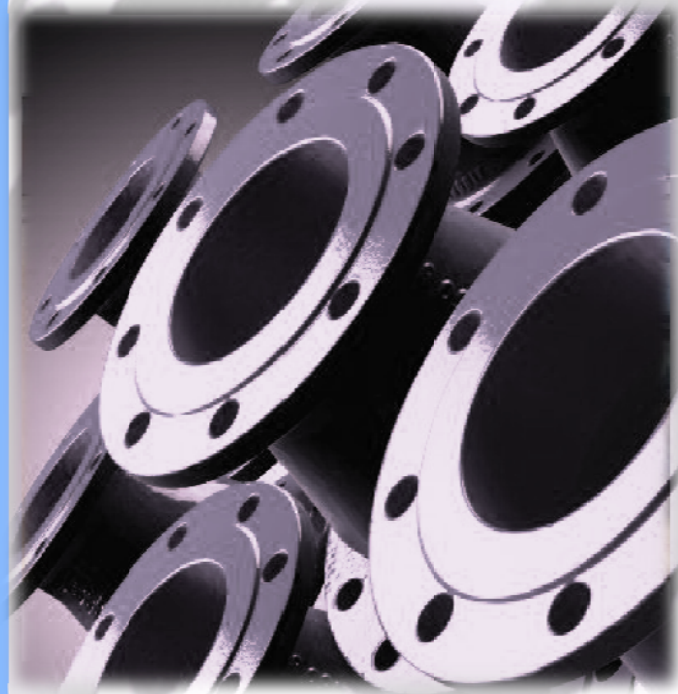
DUCTILE IRON DOUBLE FLANGE PIPE

As Per IS 8329

PRODUCT CATALOG



FOR EARTH, FOR LIFE



SANGHI TUBES PRIVATE LIMITED

79/8, LATOUCHE ROAD, KANPUR NAGAR – 208001

Tel./Mail: 7084215575, 7084225575, 9839030844/sanghipipe@gmail.com

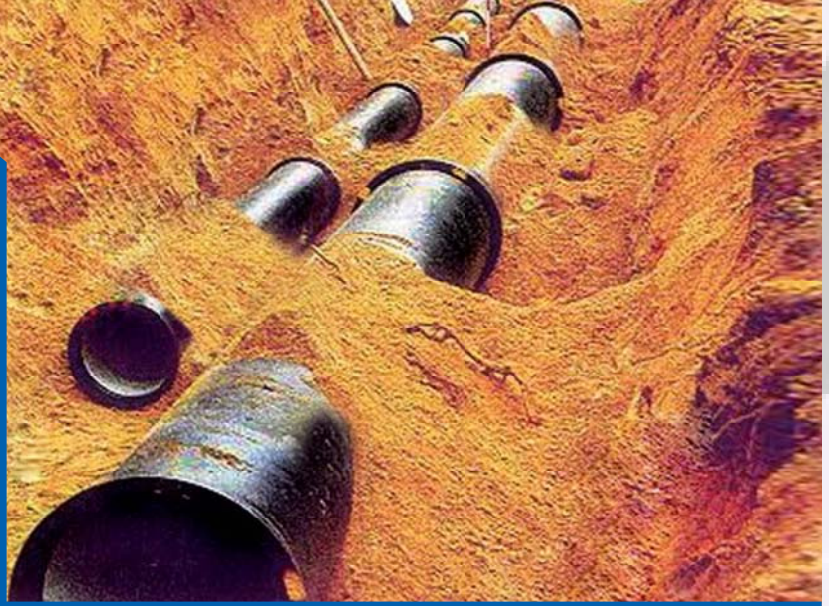
SANGHI TUBES PRIVATE LIMITED

1

Ductile Iron (DI)

In 1948, the International Nickel Co., USA developed a process in which, by inoculation of magnesium, the flaky graphite in Cast Iron (CI) changes to spheroidal nodules. This change in microstructure, resulted in spectacular increase in the ductility and consequently, tensile strength, without any detrimental effect on well known corrosion resistance of C.I. Thus, a new material called Spheroidal Graphite Cast Iron or more popularly known as Ductile Iron (DI), was born. Due to its technical superiority, the DI soon became the preferred choice of designers. This is what American Water Works Association (AWWA) says about DI, in Manual M41 - 'Ductile Iron Pipe and Fittings'.

"In addition to its greater strength, Ductile Iron pipe also possesses other superior properties of grey cast iron such as corrosion resistance, ease of casting, ease of installation, and economy. A number of studies have been conducted to determine the corrosion resistance of Ductile Iron pipe. Mostly as a result of conservative interpretation of the data, the early studies concluded that the corrosion resistance of Ductile Iron pipe was at least as good as that of grey Cast Iron pipe".



DI PRODUCTS OF SANGHI TUBES PRIVATE LIMITED

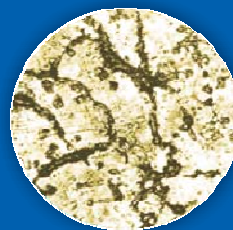
- DI Socket and Spigot (S&S) Pipes with push-on joints-Sizes DN 100-1100 mm, Class K7 & K9
- DI Barrel Pipes, Class K7 & K9 [For more information on this product please contact us]
- DI Flanged Pipes - Sizes DN 100 - 600 mm, Class K9, PN 10 (for more information please contact us)

Application of DI S&S Pipes

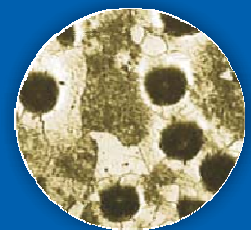
- Transmission of Raw & Potable Water
- Distribution of Potable Water
- Transmission of Domestic & Industrial Effluents
- Fire Fighting Systems
- Piling
- Ash-Slurry Handling Systems



Ductile Iron Pipes



Microstructure of Grey Cast Iron



Microstructure of Ductile Iron

Technical data of D.I. Socket & Spigot pipes

Construction Features

Ductile Iron Socket and Spigot Pipe consist of a barrel with socket at one end. The other end, which is plain, is called spigot. The Socket has a proprietary profile to house a rubber gasket manufactured to suit the profile and allow insertion of spigot upto predetermined point. Two pipes are joined by placing the rubber gasket in socket and inserting the spigot end of other pipe in socket.

Specifications

Standard : IS 8329

Size Range : DN 100-DN 1100 mm

Length : DN 100 to 300 mm - 5.5 m
DN 350 to 1100 mm - 6 m
(Pursuant to clause 15.5.1 of IS 8329, 10% of pipes in each size may be supplied in shorter lengths)

Marking : "ISI Mark" (Standard Mark) under license issued by Bureau of Indian Standards (BIS), "Kitemark" under license issued by British Standard Institute (BSI)

Class : K7 & K9

Mechanical Properties

Property	Unit	Value
Tensile Strength	Mpa	Min 420
Elongation at Break	%	Min 10
Hardness	BHN	Max 230
Density	Kg/m3	7050
Bending Strength	Mpa	>500
Impact & Crush Load	Charpy	>0.713
Bursting Strength (Minimum)	Factor of Safety against Bursting is 8-10	

Lining and Coating



Cement Mortar Lining



Zinc Coating



Bitumen Coating

Ductile iron Double Flanged Pipe for Water and Other Liquids

Flanged pipe and fittings are typical components in rigid piping systems. Such systems are particularly suited for above ground installation in the following: water filtration plants, sewage disposal plants, wastewater treatment plants, pumping stations and industrial plants, per **IS : 8329**

Flanged Pipe with Threaded Flanges

Pipe barrels conform to **IS : 8329** Flanged pipe are made by threading plain end pipe, screwing threaded flange(s) on and machine-tightening them. The bolt holes are aligned per **IS : 8329**. The plain end and the flange are then faced to insure that the flange is perpendicular to the pipe axis and the pipe/flange interface is flush. Pipe barrels and flanges have a pipe thread in accordance to IS with thread diameters adapted to Ductile Iron pipe standard outside diameters.

Threaded pipe and threaded flanges are individually fitted and the flanges are not interchangeable. D.I. Pipe only uses and recommends ductile iron flanges. The required information and the options which if desired must be specified on the purchase order for flanged pipe, such as size and finished length. *Contact us for more details.*

Advantages of DI Pipes

- Tremendous Tensile Strength
- Good Impact Resistance
- High Ductility
- Inherent Excellent Corrosion Resistance
- Reliable Internal & External Corrosion protection systems applied at factory
- High Pressure Bearing Capacity
- Flexible and Leak Tight Joints
- Easy to lay with simple push-on jointing system
- Easy to Join, Tap and Weld
- Backfilling with special material not required
- Reduced Pumping Cost
- Longer Service Life of 90 years



Flanged ductile iron pipe

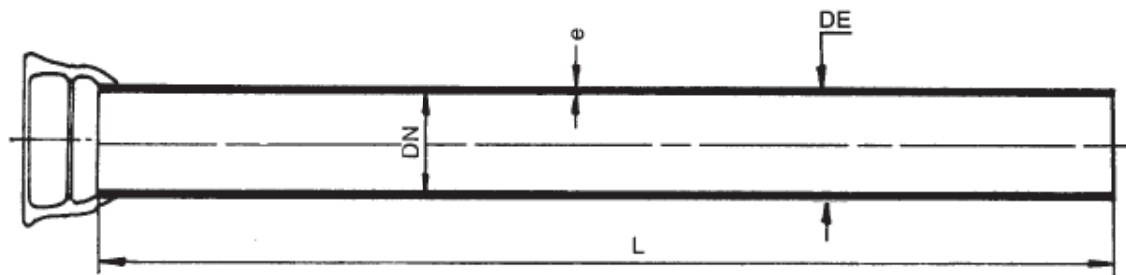
After completing flanging process each test shall be perform for quality inspection

- **Dimension test (*Flange & pipe Compatibility*)**

IS 8329 : 2000

Table 2 Dimensions of Sockets and Spigot Pipes, Classes K7, K8, K9 and K10

(Clauses 3.21, 12.2, 13.3 and 15.3.2)

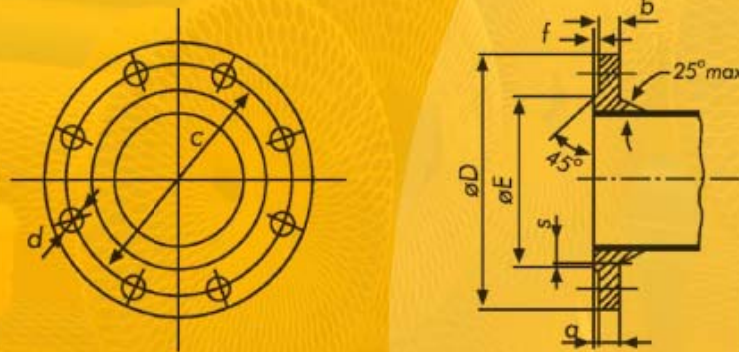


All dimensions in millimetres.

Nominal Diameter <i>DN</i>	External Diameter <i>DE</i>	Barrel Wall Thickness 'e'			
		K7	K8	K9	K10
80	98	5	6	6.0	6.0
100	118	5	6	6.0	6.1
125	144	5	6	6.0	6.3
150	170	5	6	6.0	6.5
200	222	5	6	6.3	7.0
250	274	5.3	6	6.8	7.5
300	326	5.6	6.4	7.2	8.0
350	378	6.0	6.8	7.7	8.5
400	429	6.3	7.2	8.1	9.0
450	480	6.6	7.6	8.6	9.5
500	532	7.0	8.0	9.0	10.0
600	635	7.7	8.8	9.9	11.0
700	738	9.0	9.6	10.8	12.0
750	790	9.7	10.0	11.3	12.5
800	842	10.4	10.4	11.7	13.0
900	945	11.2	11.2	12.6	14.0
1 000	1 048	12.0	12.0	13.5	15.0
1 100	1 152	14.4	14.4	14.4	16.0
1 200	1 255	15.3	15.3	15.3	17.0
1 400	1 462	17.1	17.1	17.1	19.0
1 600	1 668	18.9	18.9	18.9	21.0
1 800	1 875	20.7	20.7	20.7	23.0
2 000	2 082	22.5	22.5	22.5	25.0

NOTE — For tolerances refer to 15.

For D.I. Flange PN - 10



DIMENSIONS (Flanged Joint)

DN Nom	PN 10						PN 16					
	D Out side Dia	E Dia of raised Face	C Pitch Circle Dia	b Flange width	N No. of Bolts	Metric Bolt size/ Total length/ Thread length	D Outside Dia	E Dia of raised Face	C Pitch Circle Dia	b Flange width	N No. of Bolts	Metric Bolt size/ Total length/ Thread length
80	200	132	160	16	4	M 16 x 65/38	200	132	160	16	8	M 16 x 65/38
100	220	156	180	16	8	M 16 x 65/38	220	156	180	16	8	M 16 x 65/38
125	250	184	210	16	8	M 16 x 65/38	250	184	210	16	8	M 16 x 65/38
150	285	211	240	16	8	M 20 x 70/46	285	211	240	16	8	M 20 x 70/46
200	340	266	295	17	8	M 20 x 70/46	340	266	295	17	12	M 20 x 70/46
250	395	319	350	19	12	M 20 x 70/46	400	319	355	19	12	M 24 x 90/54
300	445	370	400	20.5	12	M 20 x 85/46	455	370	410	20.5	12	M 24 x 90/54
350	505	429	460	20.5	16	M 20 x 85/46	520	429	470	22.5	16	M 24 x 90/54
400	565	480	515	20.5	16	M 24 x 90/54	580	480	525	24	16	M 27 x 100/60
450	615	530	565	21	20	M 24 x 90/54	640	548	585	26	20	M 27 x 100/60
500	670	582	620	22.5	20	M 24 x 105/54	715	609	650	27.5	20	M 30 x 110/66
600	780	682	725	25	20	M 27 x 100/60	840	720	770	31	20	M 33 x 120/78
700	895	794	840	27.5	24	M 27 x 110/60	910	794	840	34.5	24	M 33 x 130/78
750	960	857	900	29	24	M 27 x 110/60	970	857	900	36	24	M 33 x 130/78
800	1015	901	950	30	24	M 30 x 120/66	1025	901	950	38	24	M 36 x 140/84
900	1115	1001	1050	32.5	28	M 30 x 120/66	1125	1001	1050	41	28	M 36 x 150/84
1000	1230	1112	1160	35	28	M 33 x 130/78	1255	1112	1170	45	28	M 39 x 160/103
1100	1340	1231	1270	38	28	M 33 x 130/82	1355	1218	1270	48.5	32	M 39 x 160/110
1200	1455	1328	1380	40	32	M 36 x 140/84	1485	1328	1390	52	32	M 45 x 170/115

- **Hydrostatic test** (*each pipe shall be tested for leakage*)



- **Surface / visual test**



- **Coating Systems**



- Marking



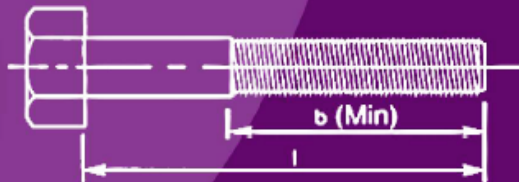
- Finished product



Accessories for Flanged Pipes

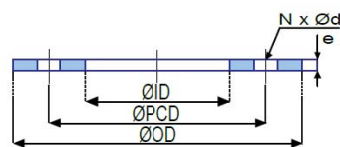
Gaskets

For Flanged pipe jointing, flat rubber gaskets as per IS 638 are used. These gaskets are made of synthetic rubber like Styrene Butadiene Rubber (SBR) or Ethylene Propylene Diethylene Monomer (EPDM). Use of poor quality gaskets made of natural rubber is not recommended.

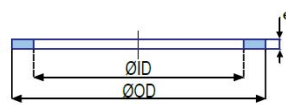


Nuts & Bolts for the Joint

Nuts and bolts: Properly galvanized nuts and bolts made of Mild Steel as per IS 1367 and whose Material Grade is 4.6 are to be used. The bolt dimension should be as per IS:1364. High tension bolts may be used where high axial load is anticipated.



Full face Gasket



Ring Type Gasket

BOLT DIMENSION

Nom. Dia. (mm)	PN 10		PN 16		PN 25		PN 40	
	No. of bolts.	Bolt size /Total length /Thread length (mm)	No. of bolts.	Bolt size /Total length /Thread length (mm)	No. of bolts.	Bolt size /Total length /Thread length (mm)	No. of bolts.	Bolt size /Total length /Thread length (mm)
80	8	M 16 x 65/38	8	M 16 x 65/38	8	M 16 x 65/38	8	M 16 x 65/38
100	8	M 16 x 65/38	8	M 16 x 65/38	8	M 20 x 70/46	8	M 20 x 70/46
125	8	M 16 x 65/38	8	M 16 x 65/38	8	M 24 x 85/54	8	M 24 x 90/54
150	8	M 20 x 70/46	8	M 20 x 70/46	8	M 24 x 90/54	8	M 24 x 100/54
200	8	M 20 x 70/46	12	M 20 x 70/46	12	M 24 x 90/54	12	M 27 x 100/60
250	12	M 20 x 70/46	12	M 24 x 90/54	12	M 27 x 100/60	12	M 30 x 120/66
300	12	M 20 x 85/46	12	M 24 x 90/54	16	M 27 x 100/60	16	M 30 x 130/72
350	16	M 20 x 85/46	16	M 24 x 90/54	16	M 30 x 110/66	16	M 33 x 140/78
400	16	M 24 x 90/54	16	M 27x100/60	16	M 33 x 120/78	16	M 36 x 150/84
450	20	M 24 x 90/54	20	M 27x 100/60	20	M 33 x 120/78	20	M 36 x 150/84
500	20	M 24 x 105/54	20	M 30 x 110/66	20	M 33 x 120/78	20	M 39 x 170/90
600	20	M 27 x 100/60	20	M 33 x 120/78	20	M 36 x 140/84	20	M 45 x 180/102
700	24	M 27 x 110/60	24	M 33 x 130/78	24	M 39 x 160/90	-	-
750	24	M 27 x 110/60	24	M 33 x 130/78	24	M 39 x 170/90	-	-
800	24	M 30 x 120/66	24	M 36 x 140/84	24	M 45 x 180/102	-	-
900	28	M 30 x 120/66	28	M 36 x 150/84	28	M 45 x 180/102	-	-
1000	28	M 33 x 130/78	28	M 39x 160/103	28	M 52 x 200/116	-	-
1100	28	M 33 x 130/82	32	M 39x 160/110	32	M 52 x 210/124	-	-
1200	32	M 36 x 140/84	32	M 45x 170/115	32	M 52 x 210/124	-	-

Jointing Procedure



Bolt Tightening sequence

Jointing Procedure for DI Flange Pipes:

- Clean the faces of the flanges.
- Inspect the gasket for any damage or manufacturing defect.
- Align the pipes properly and position the gasket between the flanges.
- Put the bolts into the flange holes one by one maintaining the alignment.
- Keep the gap between the flanges approximately uniform while tightening.
- Tighten four location bolts in order to roughly secure the adjoining flange.
- Bolts to be tightened in the correct sequence and to the appropriate torque.
- Tighten nuts to finger tight.
- Check alignment of flange faces and gasket.
- Insert remaining bolts and tighten nuts to finger tight.
- Finish with one final pass by providing torque in a clockwise direction.
- Tighten to 100% of estimated torque using the same sequence.

Installation

Precautions during installation of Flanged Pipes

- Flanged pipes are mainly for over-ground installation. Buried installation of flanged pipe is not recommended.
- Flanged joint being a rigid joint, perfect alignment of the flange face during jointing and bolt tightening is absolutely vital.
- Use of duckfoot bend at bottom of vertical flange pipe line is necessary.
- For high pressure application, even flanged pipeline (both horizontal and vertical) needs thrust block / support at bends/tees.



Applications

Over-ground and exposed installations

Ideal for over-ground pipelines and crossovers (such as on pillars, over a canal or other water bodies, roads, bridges etc.)

Vertical pipelines

Any vertical pipelines used in treatment plants, pump houses, elevated/overhead service reservoirs.

Interconnection and connection for accessories

In pump houses and water and sewage treatment plants where, various units and accessories are inter-connected.

As a Restrained Joint

Flanged joints are rigid but can act as a restrained joint, reducing the requirement of thrust blocks and external restraining devices.

As temporary installations

Applications where pipelines need to be disengaged or transferred from one location to the other.

Note: Flanged Pipes are not meant for underground installation. Due to the risk of excessive bending moments being imposed, it is recommended that the flanged pipeline is NOT buried.

“We recognize a significant opportunity to manufacture and supply water, sewer and plumbing products all over india,” said Sanjeev Sanghi, CEO of Sanghi Tubes Private Limited “It’s our next step towards bringing quality of our products to even more of the life.”